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Numerical investigation of an equivalent hydraulic aperture for rough rock fractures based on cosimulation

Weijie Zhang [a], Ziyu Peng [a], Chenghao Han [a,b,*], Shaojie Chen [b]

[a] College of Earth Science and Engineering, Shandong University of Science and Technology, Qingdao 266590, China
[b] College of Energy and Mining Engineering, Shandong University of Science and Technology, Qingdao 266590, China

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ABSTRACT

A comprehensive nonlinear seepage simulation study for rough fractures remains technically challenging due to the number of indoor tests and computational volume of numerical simulations. This study fills this gap by developing a cosimulation program to accomplish the automatic cycling of rough fracture modelling and nonlinear fracture seepage simulation. With the developed cosimulation program, 81 sets of rough fracture models are generated, and 427 sets of seepage simulations are performed. Combined with a theoretical analysis, the mechanism of the effects of roughness and fracture aperture on the flow field structure during nonlinear seepage is determined. The equivalent hydraulic aperture (EHA) is introduced as an index to quantitatively evaluate the fluid transport efficiency in the seepage process. The results show that there is a linear relationship between the fracture aperture and the EHA and a logarithmic relationship between the fracture roughness and the EHA. Finally, a regression prediction model of EHA with roughness and average fracture aperture characteristics is established through regression analysis, and a neural network prediction model of EHA is established based on a BP neural network optimized by an SSA algorithm. The developed models are verified to provide high accuracy for EHA prediction.

1. Introduction

Natural rock masses have undergone long diagenesis and complex geological processes such as tectonic evolution, and numerous structural surfaces such as joints and fractures have formed within them. Joints and fractures can significantly change the mechanical and seepage characteristics of rock masses. However, it is challenging to study the seepage process directly due to the complex fracture network structure, disordered fractures, and strong heterogeneity. Hence, accurately describing the seepage characteristics of a single fracture is a fundamental task and a popular research topic.

In initial studies, Boussinesq (1868) and Lomize (1951) simplified a rough fracture into a single flat smooth fracture and obtained a Newtonian cubic law of fluid ($Q = -h^3\nabla P/12\mu$); the channel of fluid seepage is determined by the fracture aperture h, which refers to the vertical distance between the upper and lower walls of the fracture. However, it is well known that the surfaces of rock fractures in nature are rough and tortuous (Mandelbrot, 1985; Power and Tullis, 1991; Hsiung et al., 1993). According to Tsang et al. (1983) and Crandall et al. (2010), under the influences of roughness and pressure gradients, the inertia of the fluid increases, leading to nonlinear seepage characteristics with an error of 1–2 orders of magnitude between the predicted and actual results using the cubic law. Some scholars have proposed an amendment to the cubic law to allow it to be used to describe the seepage characteristics of rough fractures (Kishida et al. 2009; Wang et al., 2015; Xu and Yu, 2022). Moreover, experiments and numerical simulations have been conducted to present and validate the empirical formula for the Forchheimer equation, which is in agreement with the nonlinear relationship of fracture seepage (Mwetulundila and Atangana, 2020; Arianfar et al., 2021). With continued research, scholars have studied the influences of roughness, anisotropy, and size on the seepage characteristics of a single fracture. Numerous research results have shown that fracture roughness significantly affects fracture seepage characteristics (Brown, 1987; Yang et al., 2001; Brush and Thomson, 2003). The effect of roughness on the seepage process in 2D fracture channels was extensively studied after Barton (1978) quantified the joint roughness coefficient (*JRC*) using 10 standard joints (Briggs, 2014; Briggs et al., 2014; Wang et al., 2017; Zhang et al., 2018). Based on the cubic law, an equivalent hydraulic aperture (EHA) was introduced to replace h, and a relationship between EHA and fracture roughness or other factors was

* Corresponding author at: College of Energy and Mining Engineering, Shandong University of Science and Technology, Qingdao 266590, China.
E-mail address: sdusthch@sdust.edu.cn (C. Han).

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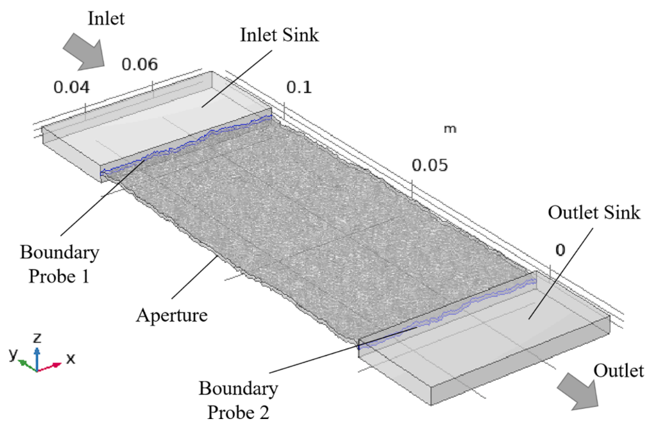

Fig. 1. Boundary settings of the numerical simulation.

established to describe the nonlinear change in fluid seepage (Wang and Su, 2002; Zhong et al., 2007; Xiao and Zhao, 2019). However, using two-dimensional parameters to describe the geometrical features of three-dimensional joints is prone errors and limitations. Therefore, a series of quantitative methods, such as the Z_{2s} descriptive, tortuosity characterization, average dip angle (Belem et al., 2000), apparent inclination standard deviation (Dang et al., 2019), and three-dimensional box methods (Xie et al., 1998; Sun and Xie, 1999), have been proposed for rough fracture surfaces under 3D conditions. These methods involve many geometrical parameters, such as the undulation angle, height distribution, and tilt morphology, and they can reasonably evaluate the roughness of a fracture surface. To date, the study of three-dimensional seepage of rough fractures has mainly been conducted by experiments on real rock samples and simulations of synthetic rock fracture models (Lanaro, 2000; Ogilvie et al., 2006; Develi and Babadagli, 2015; Rong, et al., 2017; Dang et al., 2019). While these models have all contributed to the research to a certain extent, some shortcomings remain: (1) real rock samples are insufficient in quantity to meet the number necessary for studying and repeating experiments; (2) the quantitative control of the roughness of a synthetic fracture surface is a difficult task, and it is very important to establish a suitable roughness model for a fracture surface; and (3) in a numerical simulation, it is difficult to obtain many simulation experiments because the solution takes an overly long time.

In this paper, a high-quality cosimulation program is developed to generate the surfaces of rough fractures, evaluate the geometric parameters of the fractures, and simulate the nonlinear seepage process under different fracture models. A rough fracture test device is developed, and an indoor test is conducted to verify the accuracy of the program. The nonlinear law of rough fracture seepage is explained from a microscopic perspective. Based on the developed program, 81 rough fractures are constructed, and 427 simulated cases are studied and parameterized. By applying the simulation results and theoretical equations to the EHA calculation of rough rock fractures, the EHA expression characteristics of different fractures with different properties under variable flow conditions are obtained. Finally, by considering the fracture characterization and flow parameters, an EHA prediction model with higher accuracy is constructed by using an artificial neural network (ANN) based on sparrow search algorithm (SSA) optimization.

2. Rough fracture seepage simulation method

There are three main steps to conducting a numerical simulation of rough fracture seepage: generating a rough fracture model, establishing a seepage numerical simulation model, and verifying the numerical model reliability. The construction of physical models, specific experimental process, and parameter settings are described in the following sections.

2.1. Generation and evaluation program of rough fracture surfaces

To date, there are two main methods for the generation of rough fracture surfaces: rough fracture modelling based on fractal theory (Sun and Xie, 1999; Song et al., 2021) and Fourier transform modelling based on spatial frequency distributions (Brown, 1995; Glover et al., 1998). On this basis, Ogilvie et al. (2006) combined the fractal dimension with the Fourier transform modelling method and proposed a more comprehensive mathematical model for rough fractures. Fracture geometry parameters, such as standard dimension (SD), fractal dimension (FD), mismatch length (ML), and transition length (TL), are used to generate rough fracture surfaces, in which SD controls the rise and fall of rough fracture surface heights, FD controls the decay rate of concave–convex bodies in space, and ML and TL control the degree of matching between upper and lower fracture walls. The fracture aperture distributions of generated and natural fractures are compared and analysed. It has been determined that the rough fractures generated by this method exhibit very similar surface morphologies to natural fractures (Briggs et al., 2017; Xiao et al., 2019). To evaluate the rough fracture model, the geometric parameters in Table A.1 are selected as evaluation indexes.

Based on Ogilvie et al.'s (2006) mathematical model of rough fractures and taking into account the above evaluation indicators, a rough fracture generation and evaluation program is developed with MATLAB, as follows: (1) fracture surface dimensions of 100 × 100 mm and a resolution of 512 × 512, which is a square model; (2) four key parameters are selected (ML, TL, SD, FD), and the establishment of rough fracture is initiated; (3) the coordinates of the nodes of the rough fracture are generated and saved to the TXT file; and (4) the fracture surface parameters (h_m, JRC_m, Z_{2s}, R_s, σ_θ) are calculated according to the coordinates and saved in an XLSX file, in which JRC_m, Z_{2s}, and σ_θ take the mean value of the upper and lower surfaces, shown in Table A.1.

2.2. Finite element model (FEM) of fracture seepage

Based on the above method, the finite element model of rough fracture seepage is established by using COMSOL Multiphysics numerical simulation software. Since Darcy's equation cannot describe the nonlinear seepage characteristics, the Navier-Stokes equation (N-S equation), which is more established in single-phase fluid flow, is adopted as the control equation for fluid flow (Crandall et al., 2010; Zhang et al., 2017; Xiao and Zhao, 2019; Xing, et al., 2021). For thermostatic incompressible single Newtonian fluids, the NSE can be written as follows:

$$\rho(u\cdot\nabla)u = \nabla\cdot[-pI + K] + F \tag{1}$$

$$\rho\nabla u = 0 \tag{2}$$

where $\mathbf{u}$ (m^3/s) is the fluid velocity, p (Pa) is the fluid pressure, ρ (kg/m^3) is the fluid density, μ (Pa s) is the hydrodynamic viscosity, $\mathbf{I}$ (N/m^2) is the unit tensor, and $\mathbf{F}$ (N) is the volume force. Eqs. (1) and (2) are momentum conservation equations and mass conservation equations, respectively. Since the hypothesis ignores the gravitational effect, the gravitational term is omitted from Eq. (1).

Additionally, to simulate the vortex flow of a fluid affected by the roughness of the fracture surface, the standard k–ε model based on the transport equation of turbulent kinetic energy k and its dissipation rate ε is adopted (Schrauf and Evans, 1986; Zou et al., 2015; Briggs et al., 2017). The binding equations of turbulent kinetic energy k and turbulent dissipation rate ε are as follows:

$$\rho(u\cdot\nabla)k = \nabla\cdot\left[\left(\mu + \frac{\mu_T}{\sigma_k}\right)\nabla k\right] + P_k - \rho\varepsilon \tag{3}$$

$$\rho(u\cdot\nabla)\varepsilon = \nabla\cdot\left[\left(\mu + \frac{\mu_T}{\sigma_k}\right)\nabla\varepsilon\right] + C_{\varepsilon 1}\frac{\varepsilon}{k}P_k - C_{\varepsilon 2}\rho\frac{\varepsilon^2}{k} \tag{4}$$

where k is the turbulent kinetic energy, which can be expressed as

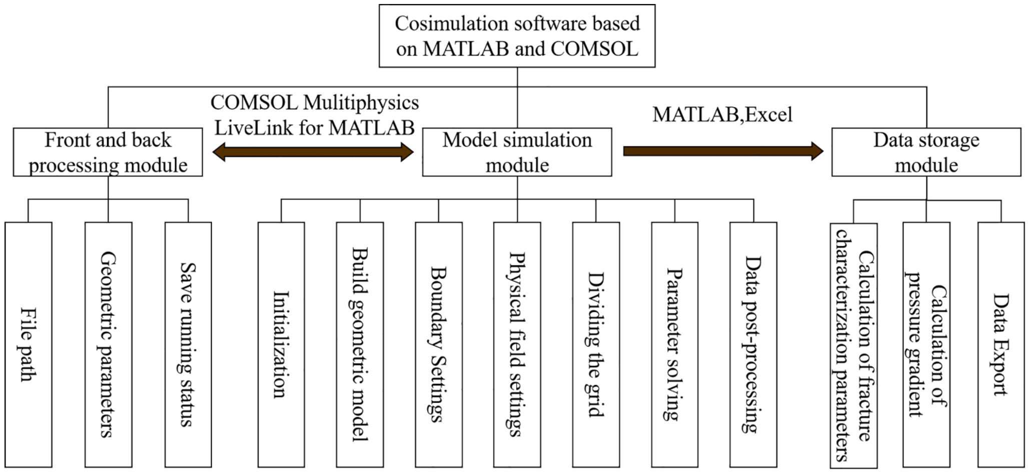

Fig. 2. Architecture diagram of the cosimulation program based on MATLAB and COMSOL.

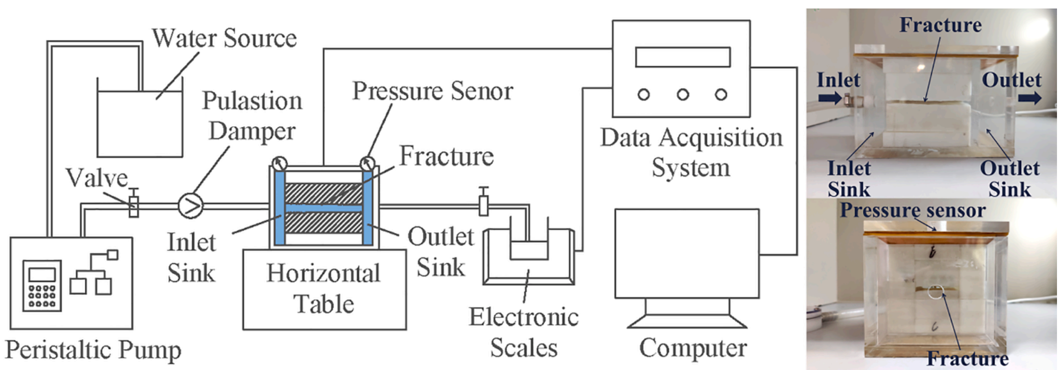

Fig. 3. Schematic view of the seepage test setup.

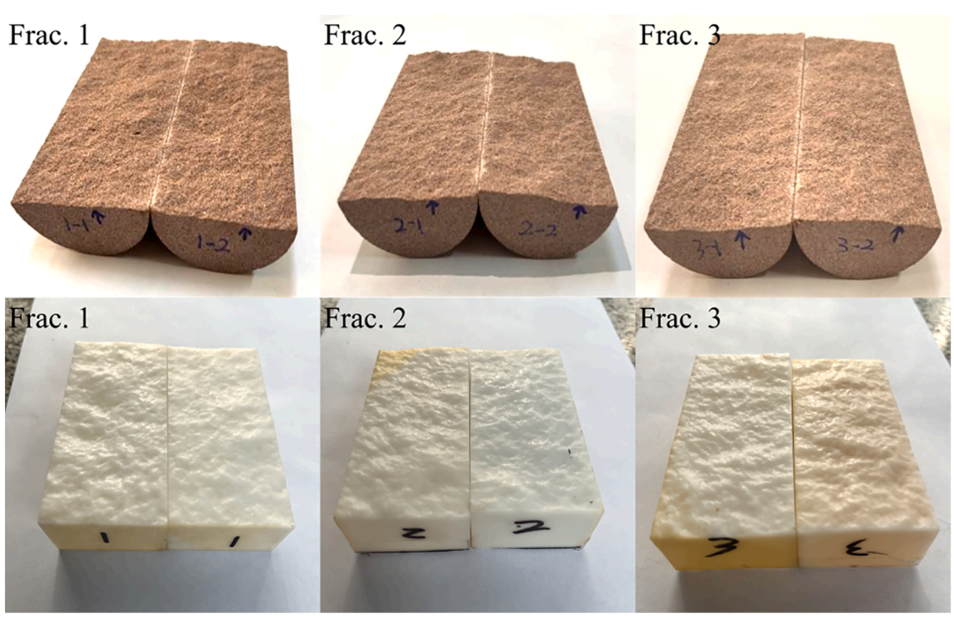

Fig. 4. Morphology and 3D printing model of the upper and lower walls of fractures.

follows:

$$k = (\mu + \mu_{\mathrm{T}})\left(\nabla u + (\nabla u)^{\mathrm{T}}\right) \tag{5}$$

where P_k is a turbulent kinetic energy generator and can be expressed as follows:

$$P_{\mathrm{k}} = \mu_{\mathrm{T}}\left[\nabla u \cdot \left(\nabla u + (\nabla u)^{\mathrm{T}}\right)\right] \tag{6}$$

where μ is the fluid viscosity, μ_{T} is the turbulent viscosity and $\mu_{\mathrm{T}} =$

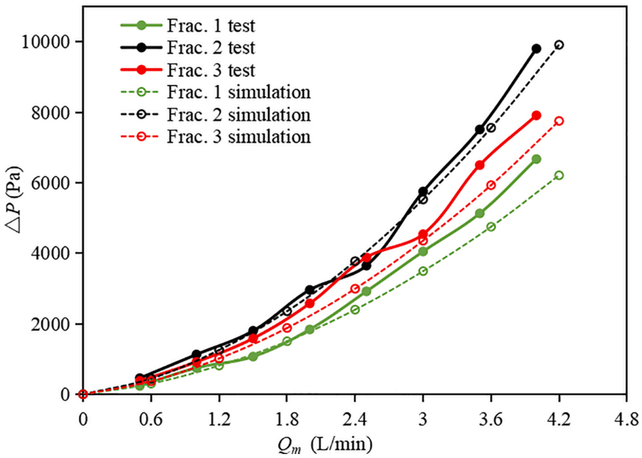

Fig. 5. Relationship between the seepage velocity and the pressure difference.

$\rho C_\mu k^2/\varepsilon$. The turbulent model parameters take default values: $C_{\varepsilon 1} = 1.44$; $C_{\varepsilon 2} = 1.92$; $C_\mu = 0.09$. The Planck constants of k and ε are $\sigma_k = 1$, $\sigma_\varepsilon = 1.3$, and $k_v = 0.41$. Last, $B = 5.2$.

In conclusion, after the seepage model of a rough fracture is established, a finite element simulation can be established by combining Eqs. (1)–(4) to analyse and calculate its nonlinear seepage characteristics.

A rough fracture seepage model is presented in Fig. 1. MATLAB is programmed to process the resulting rough fractures to create rectangular fracture channels measuring 50 × 100 mm, as the rectangular shape is more general (Crandall et al., 2010; Zhang and Nemcik, 2013; Rong et al., 2017). Sinks are installed on both sides of the crevasse, and the sizes of the sinks are chosen to ensure that both ends of the fracture are included to reduce the boundary effect and make it easier to obtain convergent solutions; the sink sizes are the same size in the same model. For the boundary condition setting, the inlet section is defined as a constant mass velocity, and the outlet end is defined as an open boundary. Assuming that the direction of flow is along the length, the other boundaries are impermeable; the coupling effect between fluid flow and rock is not considered. Boundary Probe 1 and Boundary Probe 2 are arranged at both ends of the rough fracture model to monitor the pressure values at the entrance and exit, respectively. When the seepage reaches a stable state, the difference between Boundary Probe 1 and Boundary Probe 2 is the difference in pressure before and after the seepage of the rough fracture.

Solving a nonlinear seepage process by numerical simulation typically takes 5 ~ 6 h. When a great number of numerical models are

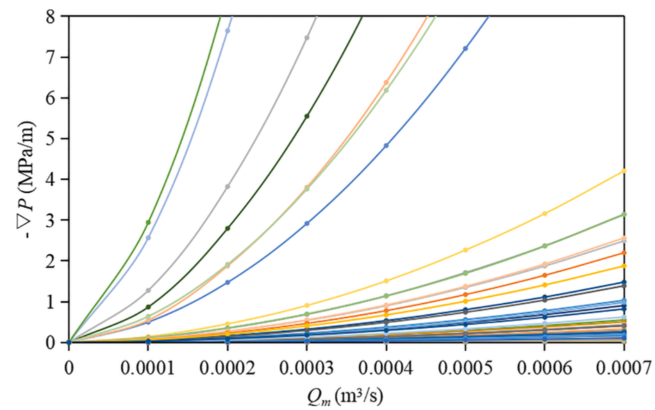

Fig. 7. Relationship between the pressure gradient and flow of water in the rough fracture.

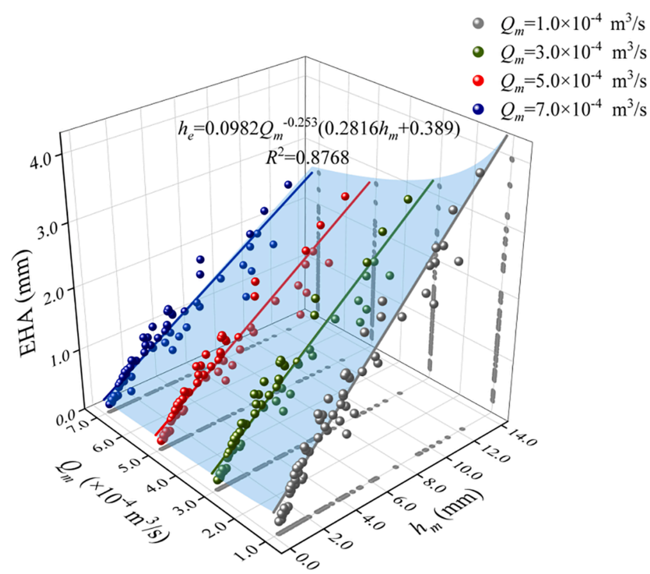

Fig. 8. Relationship between the fracture aperture and the equivalent hydraulic aperture.

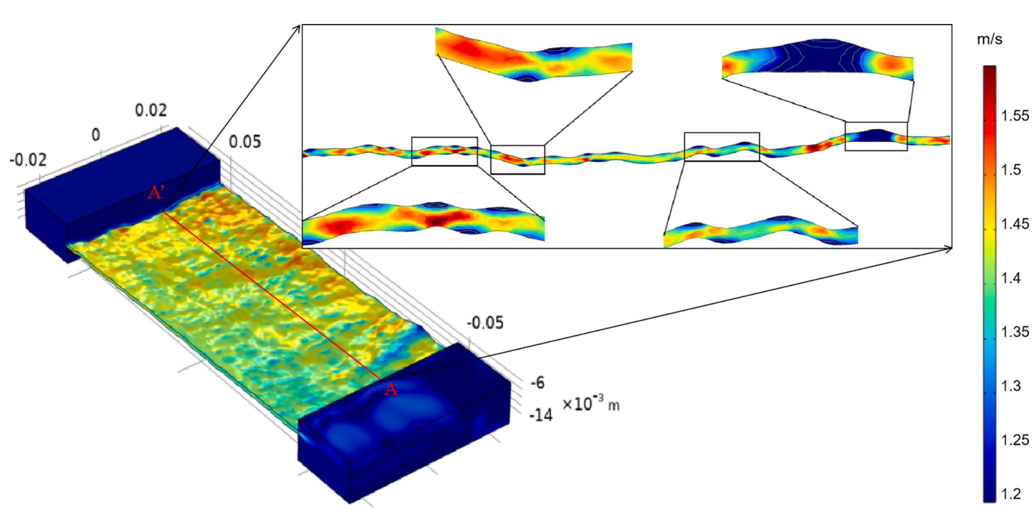

Fig. 6. Distribution of the fluid velocity and pressure in the rough fracture.

Table 1
Fitting results under different volumetric velocities.

Q_m (m3/s)	Fit EHA		Fit C_h	
	EHA	R2	C_h	R2
0.0001	$h_e = 0.2816h_m + 0.3890$	0.8958	$C_h = -0.142\ln(Z_{2s}) + 0.5225$	0.6529
0.0003	$h_e = 0.212h_m + 0.3085$	0.8815	$C_h = -0.113\ln(Z_{2s}) + 0.4065$	0.6017
0.0005	$h_e = 0.1824h_m + 0.2755$	0.8622	$C_h = -0.1\ln(Z_{2s}) + 0.3562$	0.5811
0.0007	$h_e = 0.1659h_m + 0.2453$	0.8443	$C_h = -0.092\ln(Z_{2s}) + 0.3223$	0.5351

simulated, the calculation time is thousands of hours. High labour costs and computational loads make the work challenging. To improve the simulation efficiency and realize many simulations, a cosimulation program based on MATLAB and COMSOL is developed by combining the rough fracture generation and evaluation program and finite element seepage model. The program uses MATLAB to set parameters to generate a fracture model and calls on COMSOL Multiphysics with MATLAB to complete the seepage simulation, data computation, and storage of the model. The cosimulation program consists of three modules:

(1) Front and back processing module: used to define the parameter list of the model and save the program running state.
(2) Model simulation module: data are transmitted to front and back processing modules using COMSOL's MATLAB LiveLink interface, and the flow module of COMSOL is called upon to simulate and save the results of the calculations after the model parameters are obtained.
(3) Data storage module: TXT files for the bulk import of models, the fracture characterization parameters for each model and the pressure gradient between the two boundary probes are calculated, and the results are saved in Excel files.

Using the program, model parameters (ML, TL, SD, FD, Q_m) are simply saved in the corresponding array format. After opening the loop, MATLAB automatically invokes the array and generates a fracture model to be saved as a TXT file. COMSOL automatically invokes the TXT file to create the corresponding analogue boundary conditions for simulation. After the simulation is over, the data are transferred to the MATLAB workspace, and the data are processed and saved in sequence until the set array traversal is complete. Fig. 2 shows the program architecture.

2.3. Validation

To further study the nonlinear seepage law of rough fractures by using the proposed numerical simulation model, a rough fracture seepage test device is developed to verify the numerical simulation results.

The device mainly includes a water source, pressure sensor, injection system, rectifier, closed fracture seepage chamber, and data acquisition device. The injection system is a peristaltic pump with a flow rate of 0–6000 mL/min, which can be used for larger flow velocity fracture seepage tests. The rectifier is a fluid pulsation damper that can effectively correct the liquid flow out of the peristaltic pump and keep steady flow out. The seepage chamber size of the closed fracture is 140 × 100 × 100 mm, and the rest are filled with 3D-printed cuboid models to accommodate the different fracture sizes and apertures. The specific device is shown in Fig. 3.

The specific test steps are as follows. (1) The sandstone specimen

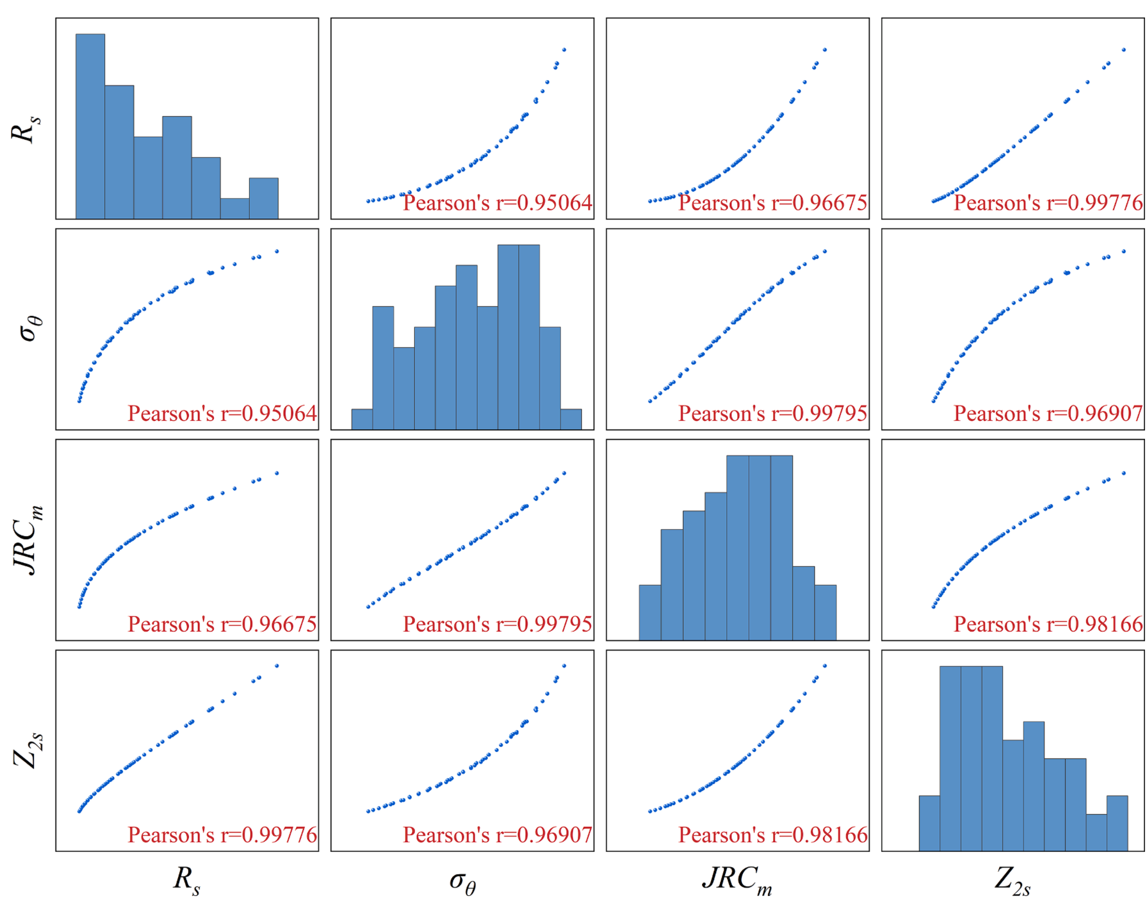

Fig. 9. Scatter matrix of the fracture parameters.

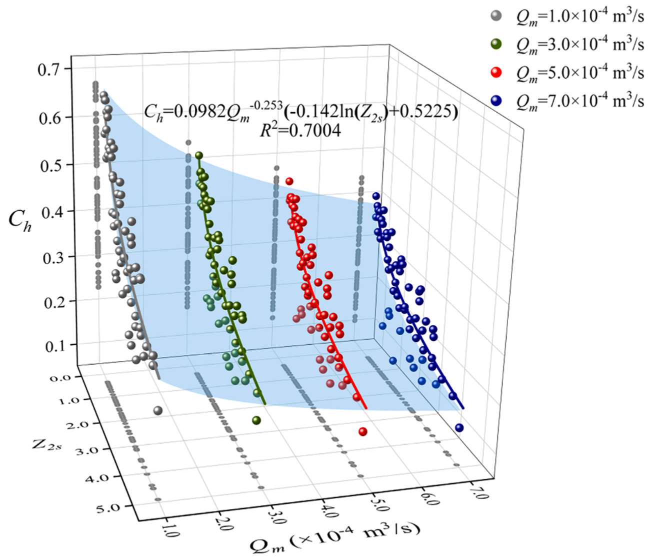

Fig. 10. Relationship between the fracture roughness and the equivalent hydraulic aperture conversion coefficient.

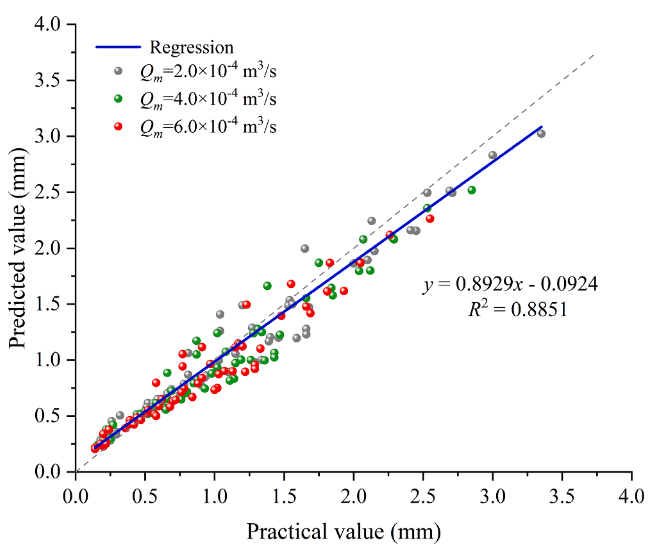

Fig. 11. Relationship between the practical and predicted equivalent hydraulic apertures.

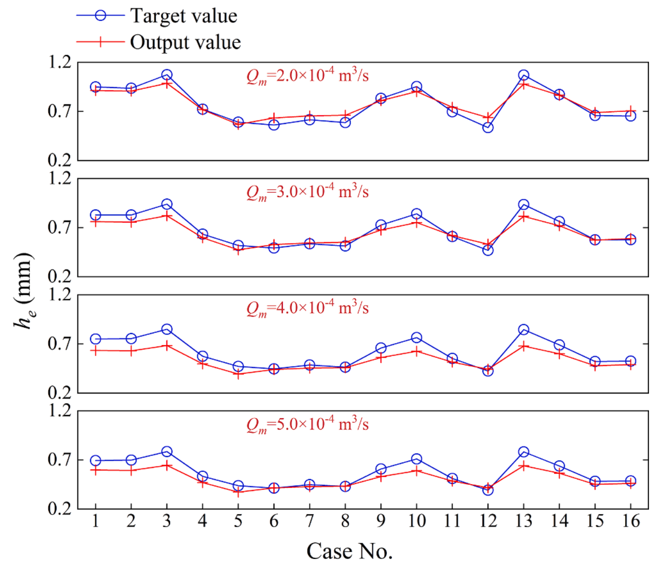

Fig. 12. Validation of the prediction model for calculating EHA based on the regression prediction model.

Table 2
Parameter selection of the optimization algorithm.

Algorithm	Parameters				
GA–BP	Population	Crossover rate	Mutation rate	Error value	Iterations (N)
	30	0.2	0.1	10^{-4}	100
SSA–BP	population	discoverers	vigilantes	safety value (S)	iterations (N)
	30	0.2	0.1	0.8	100

material is prepared into cylinders with dimensions of 50 mm in diameter and 100 mm in length, and the displacement of the specimen according to a Brazilian splitting control method causes the specimen to split (Goodman, 1976), forming a penetrating fracture, as shown in Fig. 4. Frac. 1 denotes fracture specimen 1, Frac. 1–1 denotes the upper wall, and Frac. 1–2 denotes the lower wall. (2) The fracture surface topography data are obtained by a 3D laser scanner. The model with rough fracture surfaces is constructed by modelling software. Finally, the single rough fracture physical model is made by 3D printing technology, with dimensions of 100 mm in length, 50 mm in width, and 30 mm in thickness, as shown in Fig. 4. (3) The upper surface and the lower surface of the fracture are combined. The lower surface is controlled to remain unchanged, and the upper surface is raised by 1 mm. This generation process is similar to natural rock fractures and the sample preparation procedures in experiments (Tatone and Grasselli, 2012; Chen et al., 2015; Xing, et al., 2021). (4) The seepage test of the rough fracture model under a volume velocity of 0–4.2 L/min (0–7 × 10^{-5} m^3/s) is performed to the extent permitted by the test device. (5) A corresponding finite element model for numerical simulation is constructed that records the pressure values of all simulated inlets and outlets.

Because of the low pressure of water flow in the fracture channel, the deformation of the rock replica can be ignored. Moreover, the absorbance of the photosensitive resin material is low. The application of 3D printing technology can effectively solve the problem of nonrepeatable experiments on rough fracture surfaces.

Fig. 5 shows the variation in the pressure difference between the seepage inlet and outlet with volume velocities under different rough fracture models. With the increase in the volume flow velocity, the pressure difference shows nonlinear increases, and the pressure difference change rate increases under the same volume flow velocity difference. When comparing the experimental and numerical results of different models, the relative error is determined to be 11.40%, 7.01%, and 15.13%. The mean error between the numerical simulation results and the experimental results (test) is found to be 11.18%. Typically, the error between the numerical simulation results and the experimental (test) results is considered acceptable when within ± 15% (Xiong et al., 2018; Xiong et al., 2021). The finite element seepage model can describe the nonlinear process of seepage well and verify the reliability of the seepage finite element model.

Since it is difficult to capture the microfluidic flow process in indoor experiments, the seepage process in the rough fracture is analysed by numerical simulation. Fig. 6 shows the velocity field distribution of the Frac. 1 fracture simulation under the condition of Q_m = 4.2 L/min (7 × 10^{-5} m^3/s), which shows the velocity distribution of the cross-section of A–A’. As shown in Fig. 6, the inlet velocity is stable and uniform, indicating its function of stabilizing the inlet flow into rough rock fractures. During the seepage process, the existence of a concave–convex body on the surface of the fracture prevents the flow of fluid, which causes the

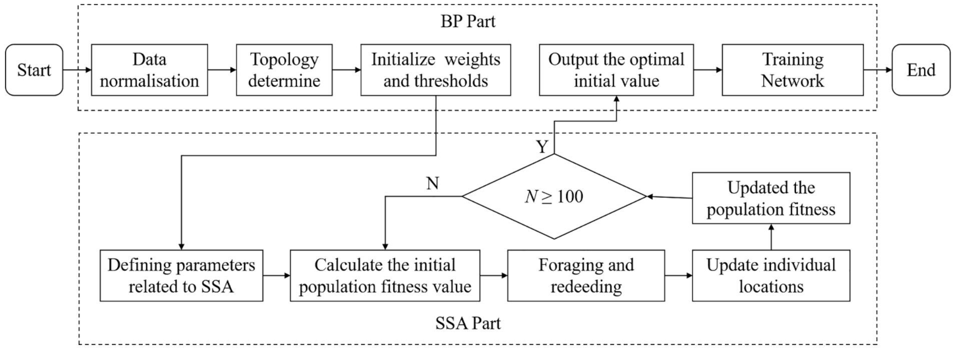

Fig. 13. Calculation flow chart of the SSA–BP neural network.

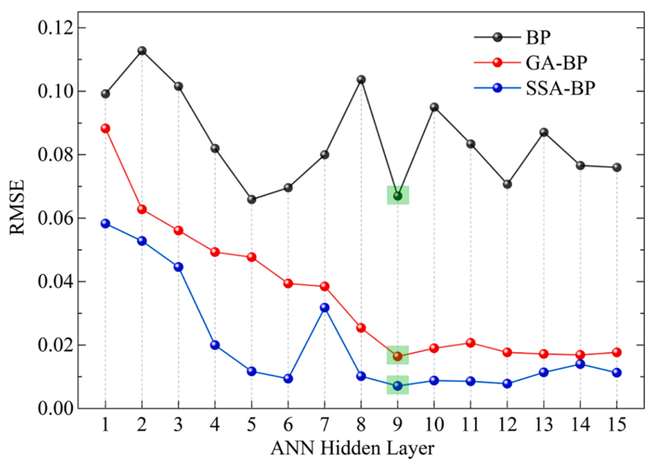

Fig. 14. Comparison chart of the RMSEs of neural networks.

flow rate and direction of fluid near the concave–convex body to change markedly. A similar flow field characteristic is obtained by simulating groups of different rough fractures. The flow velocity is mostly lower in the fracture bend and where the fracture aperture changes from narrow to wide; the flow velocity is mostly higher in the middle of the fracture and where the fracture aperture changes from wide to narrow. The results show that the variations in the fracture roughness and aperture are the main reasons for complex flow phenomena, such as vortex flow or backflow. As the volume velocity increases, the effective horizontal flow area narrows inertia increases, and the internal friction resistance and local energy loss of the fluid increase, eventually causing nonlinear seepage behaviour in the rough fracture. Additionally, the high velocity is concentrated in the middle of the fracture, and the low velocity near the fracture wall is especially in the high bending direction. It can be concluded that the fracture aperture does not fully contribute to the transport of fluid due to the roughness of the fracture wall. The effective transport of fluid is concentrated only in the central part of the fracture, and the EHA is proposed for evaluation of this phenomenon, which is closely related to fracture aperture and fracture roughness under the same injection flow (Wang et al., 2017; Tsang and Witherspoon, 1983; Dang et al., 2019).

3. Evaluation and prediction of the EHA

The finite element model of fracture seepage has been successfully constructed and proven to be effective in simulating the characteristics of nonlinear seepage. To explore the quantitative relationship between the EHA and fracture parameters, an orthogonal experiment is designed, and a large-scale simulation experiment is conducted based on the previous cosimulation program. More details are provided in the following sections.

3.1. Orthogonal design

An orthogonal experiment was conducted by using the finite element mathematical model constructed in Section 2. Four main fracture geometric characteristic parameters were selected, including ML, TL, SD, and FD. Nine levels were selected for each parameter, and 81 rough fracture models were constructed, as shown in Table A.2.

Models were numerically seeped at intervals of 1×10^{-4} m^3/s, with a volume flow Q_m of 1×10^{-4} to 7×10^{-4} m^3/s at the entrance, and 7 constant seepage simulations were performed for each model, totalling $81 \times 7 = 567$ sets of simulations. The fracture characterization parameters of each model were calculated, including h_m, JRC_m, Z_{2s}, R_s, and

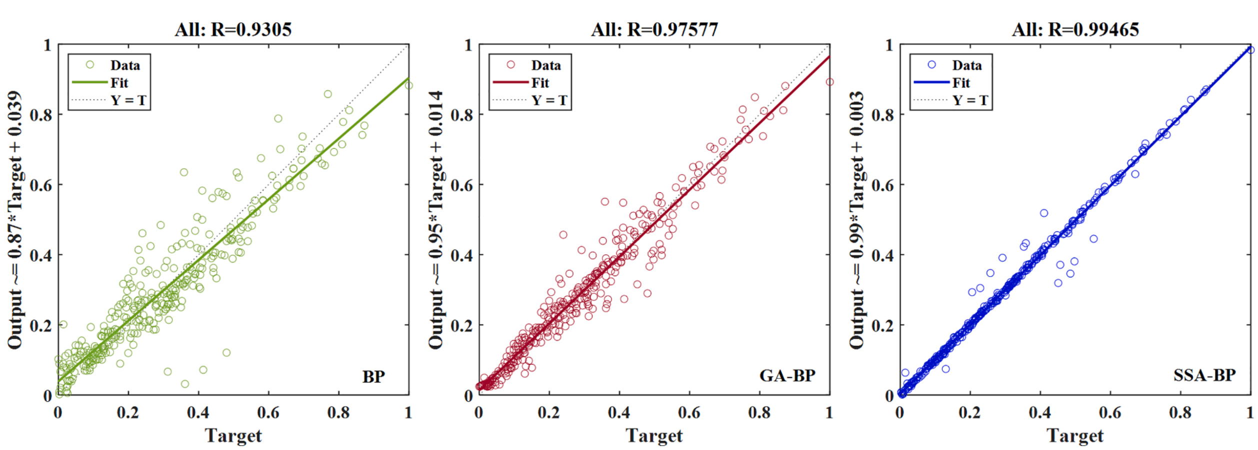

Fig. 15. Correlation coefficient (R) of the predictive model developed based on ANN: target (T) is the real EHA, and output (Y) is the predicted EHA.

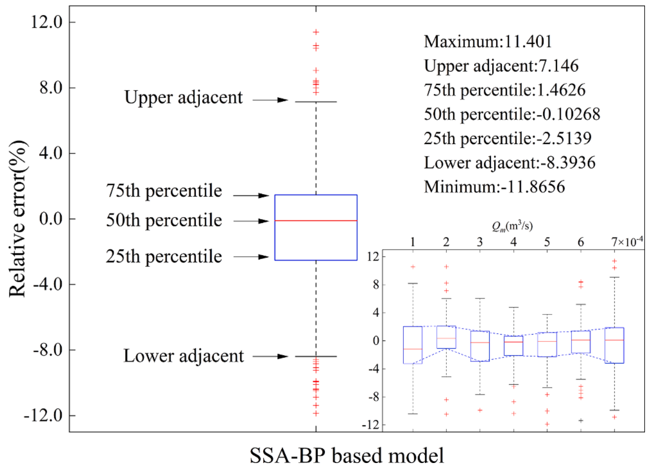

Fig. 16. Box plot of the relative error between the real and predicted EHAs: Relative Error = 100*(Output-Target)/Target.

σ_θ. The hydraulic aperture (h_e) was calculated based on the surrogate formula of the Newtonian cubic law of fluid in Eq. (7) obtained at the inlet and outlet, which in this simulation was equivalent to the EHA.

$$h_e = \sqrt[3]{\frac{12\mu Q_m}{\rho W \nabla P}} \tag{7}$$

where μ and ρ are the viscosity and density of the fluid, respectively, W is the width of the fracture channel, Q_m is the volume velocity, and ∇P is the inlet and outlet pressure gradient. Table A.2 shows the fracture-generating parameters and the roughness characterization parameters of each scheme. The final set of 427 valid test results obtained by the program are shown in Table A.3.

3.2. Development of prediction models for EHA

3.2.1. Regression-based model

Fig. 7 shows the variation in -∇P with Q_m under different experimental scenarios. This change shows that the method can describe the nonlinear characteristics of the fluid. Many studies have shown that the relationship between the pressure gradient ∇P and volume flow rate Q_m can be described using the Forchheimer formula ($-\nabla P = AQ + BQ^2$). To describe the nonlinear trend of the seepage process, all test results have been fitted in this section and are shown in Table A.3.

Based on data in Table A.3, Fig. 8 shows the relationship between fracture aperture h_m and EHA at different Q_m. In the simulated volume flow rate range, the EHA is significantly lower than h_m during seepage. As Q_m increases, the inertia of the fluid becomes more obvious, and the EHA decreases. The EHA formula under the corresponding Q_m is fitted by a linear relation, and the result is shown in Table 1. Both Q_m and h_m are selected as independent variables, and the custom function $y = ax_1^b(cx_2 + d)$ is set to fit the relationship between h_m and EHA at different Q_m values (Eq. (8)). $R^2 = 0.8768$ for Eq. (8). The two can be expressed linearly, but if only the average fracture aperture h_m is considered, the calculation results produce great errors, and the error increases with increasing h_m.

$$h_e = 0.0982Q_m^{-0.253}(0.2816h_m + 0.389) \tag{8}$$

As described in Section 2.1, R_s, σ_θ, JRC_m, and Z_{2s} are introduced to quantify the roughness of the fracture. However, it is unadvisable to take too many elements representing the same parameters when building an

Table A1
Fracture roughness factors in the literature.

Methods	Expressions	Highlights/limitations/applications	References
Mechanical fracture aperture h_m	$h_m = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}\left(z_{i,j}^t - z_{i,j}^b\right)$ where $z_{i,j}^t$ and $z_{i,j}^b$ are points on the top and bottom of the (*i*,*j*) coordinates, respectively; m and n are the number of nodes in the x and y directions, respectively.	Mechanical fracture aperture refers to the vertical distance between the upper and lower walls of a fracture, which is used to indicate the fracture opening.	Belem et al., 2000
Joint roughness coefficient *JRC*	$Z_2 = \frac{1}{L}\sqrt{\int_0^L\left(\frac{dz}{dx}\right)^2 dx}$ where L is the total length of the upper profile along the x direction, x is the sampling interval, and z is the fluctuation difference between the two adjacent nodes. $JRC = 51.16Z_2^{0.531} - 11.44$ $JRC_m = \frac{1}{N}\sum_{i=1}^{N} JRC$ N is the number of nodes in the y directions.	*JRC* can be expressed by the expression $JRC = a(Z_2)^b + c$, in which a, b and c are associated with the sampling interval of the profile. Z_2 is the square root of the slope between adjacent points. In this paper, *JRC* is estimated by a sampling interval of 0.5 mm. The roughness of the fracture surface is characterized by the mean *JRC*.	Ge et al., 2012
Z_{2s} descriptive method	$Z_{2s} = \left(\frac{1}{(N_x-1)(N_y-1)}\left[\frac{1}{\Delta x^2}\sum_{i=1}^{N_x-1}\sum_{i=1}^{N_y-1}\frac{(z_{i+1,j+1}-z_{i,j+1})^2+(z_{i+1,j}-z_{i,j})^2}{2} + \frac{1}{\Delta y^2}\sum_{i=1}^{N_y-1}\sum_{i=1}^{N_x-1}\frac{(z_{i+1,j+1}-z_{i+1,j})^2+(z_{i,j+1}-z_{i,j})^2}{2}\right]\right)^{\frac{1}{2}}$ where N_x and N_y are the numbers of nodes in the x and y directions, respectively, and Δx and Δy are the sampling intervals in the x and y directions, respectively.	Z_{2s} description is an extension of the 2D Z_2 description. Assuming that the points on the surface of a three-dimensional fracture structure are continuously microscopic and sampling intervals are small.	Belem et al., 2000
Tortuosity characterization R_s	$R_s = \frac{1}{2}(R_{top} + R_{bottom}) = \frac{1}{2}\left(\frac{S_{top}}{S'_{top}} + \frac{S_{bottom}}{S'_{bottom}}\right)$ where R_{top} and R_{bottom} are the R_s values on the top and bottom of the fracture, respectively.	R_s represents the ratio of the actual area S of the fracture surface to the projected area S' of the fracture surface. The roughness of the fracture can be expressed as the average of the R_s values of the upper and lower surfaces.	Belem et al., 2000
Apparent inclination standard deviation σ_θ	The planar normal $\boldsymbol{n}$=(a,b,c) with a triangular element ABC is as follows: $n = AB \times AC$ The normal $\boldsymbol{n}$ is projected as $\boldsymbol{n}_1$=(a,0,c) on the shear plane (0,1,0). The normal's angle to shear vectors (0,0,1) is α: $\alpha = (n_1 \cdot s)/(\lvert n_1\rvert \lvert s\rvert)$ The dip angle θ can be written as follows: $\sin\theta = \cos\langle n, n_1\rangle$ the apparent inclination angle θ^* can be obtained as follows: $\tan\theta^* = -\tan\theta\cos\alpha$ σ_θ can be calculated according to the data set of θ^*	In terms of the slope of the fracture surface, mean inclination θ_s is used for characterizing the roughness of the surface. Dang (2019) found that the mean inclination of natural rock formations is approximately 0°, leading to a θ_s value that may not be a valid evaluation parameter. However, the standard deviation of the angle of inclination σ_θ can accurately reflect the roughness of the rough surface.	Dang et al., 2019

Table A2
Fracture data and fracture roughness evaluation parameters.

No.	ML	TL	SD	FD	h_m (mm)	R_s	σ_θ	JRC_m	Z_{2s}
1	5	20	0.99	2.5	1.35	1.259	18.831	10.913	0.797
2	8	90	2.31	2.75	12.32	3.321	49.013	34.894	3.533
3	5	50	0.33	2.7	0.94	1.092	11.454	4.804	0.447
4	1	40	0.66	2.7	0.15	1.314	21.003	12.313	0.894
5	1	20	2.31	2.85	0.69	4.073	53.771	39.936	4.428
6	6	30	0.99	2.55	1.89	1.324	21.082	12.54	0.911
7	6	60	0.33	2.75	1.21	1.115	12.854	5.976	0.505
8	6	20	1.98	2.65	3.91	2.371	40.36	27.002	2.365
9	7	50	2.97	2.5	8.88	2.395	40.304	27.194	2.391
10	7	40	0.99	2.75	3.8	1.723	31.053	19.571	1.514
11	5	40	1.32	2.8	4.16	2.292	39.71	26.242	2.267
12	5	80	0.66	2.9	2.99	1.643	29.738	18.421	1.402
13	3	10	2.31	2.8	4.16	3.685	51.531	37.437	3.967
14	1	10	0.33	2.5	0.02	1.034	6.858	0.367	0.266
15	8	20	0.66	2.6	1.44	1.203	16.838	9.301	0.693
16	7	90	1.32	2.6	6.03	1.633	28.959	18.224	1.385
17	7	70	0.33	2.8	1.49	1.142	14.329	7.151	0.567
18	3	30	0.33	2.6	0.45	1.057	8.931	2.513	0.347
19	2	30	2.31	2.9	4.36	4.48	55.864	42.388	4.907
20	5	30	2.97	2.85	8.54	5.154	58.543	46.082	5.693
21	2	40	1.65	2.85	3.02	3.016	46.957	32.607	3.163
22	1	90	1.98	2.55	0.5	1.951	34.589	22.478	1.821
23	6	70	2.64	2.7	10.16	3.36	49.192	35.159	3.576
24	4	40	0.33	2.65	0.61	1.073	10.137	3.641	0.394
25	8	30	2.64	2.8	9.32	4.166	54.2	40.486	4.533
26	8	70	1.32	2.5	5.32	1.417	23.696	14.517	1.063
27	4	50	2.31	2.55	3.49	2.184	37.865	25.078	2.125
28	6	90	0.66	2.8	2.7	1.462	25.398	15.399	1.134
29	9	90	0.33	2.9	1.8	1.207	17.384	9.447	0.701
30	8	40	1.98	2.9	8.26	3.883	52.84	38.759	4.207
31	6	40	2.31	2.5	3.96	1.981	34.89	22.817	1.86
32	3	40	2.64	2.55	3.15	2.423	40.741	27.471	2.428
33	8	10	1.65	2.7	4.14	2.267	39.212	25.982	2.235
34	9	70	2.31	2.65	10.66	2.688	43.668	29.882	2.759
35	1	80	2.97	2.65	0.84	3.339	49.046	34.988	3.548
36	3	80	1.98	2.5	2.61	1.783	31.665	20.345	1.594
37	2	50	0.66	2.75	0.93	1.383	23.193	13.856	1.009
38	4	60	1.32	2.9	4.25	2.722	44.484	30.211	2.805
39	4	70	0.66	2.85	2.6	1.549	27.605	16.923	1.265
40	4	30	1.98	2.7	4.09	2.625	43.168	29.337	2.681
41	9	60	0.99	2.7	4.22	1.602	28.516	17.763	1.341
42	7	80	2.31	2.7	9.67	2.988	46.395	32.374	3.129
43	9	80	1.32	2.55	6.37	1.516	26.292	16.331	1.214
44	5	60	2.31	2.6	5.24	2.42	40.766	27.45	2.425
45	9	10	2.64	2.85	9.11	4.612	56.382	43.128	5.06
46	8	50	0.99	2.65	3.2	1.494	25.967	15.977	1.182
47	3	90	0.99	2.85	2.86	2.006	35.962	23.169	1.897
48	3	50	1.65	2.9	4.43	3.299	49.204	34.75	3.506
49	8	80	0.33	2.85	1.9	1.172	15.847	8.31	0.633
50	7	10	0.66	2.55	1.03	1.161	14.943	7.865	0.607
51	2	80	0.99	2.8	1.66	1.86	33.583	21.384	1.7
52	5	90	2.64	2.65	9.73	3.011	46.511	32.531	3.153
53	3	60	0.66	2.65	1.01	1.254	18.869	10.793	0.788
54	6	10	2.97	2.9	9.33	5.683	60.448	48.789	6.309
55	4	90	1.65	2.5	4.27	1.594	27.962	17.602	1.328
56	7	30	1.65	2.65	3.92	2.063	36.399	23.802	1.971
57	6	50	1.32	2.85	4.81	2.5	42.156	28.25	2.53
58	7	60	1.98	2.85	9.17	3.541	50.743	36.461	3.795
59	8	60	2.97	2.55	11.26	2.668	43.267	29.688	2.732
60	9	40	2.97	2.6	9.66	2.982	46.177	32.289	3.117
61	4	20	2.97	2.8	7.3	4.647	56.425	43.319	5.1
62	3	20	1.32	2.7	2.08	1.925	34.471	22.184	1.788
63	2	10	1.32	2.65	1.26	1.769	31.713	20.185	1.577
64	7	20	2.64	2.9	9.15	5.079	58.378	45.71	5.609
65	2	70	1.98	2.6	1.8	2.148	37.475	24.708	2.078
66	3	70	2.97	2.75	6.21	4.175	54.115	40.535	4.542
67	6	80	1.65	2.6	6.32	1.883	33.564	21.661	1.732
68	1	50	2.64	2.6	0.45	2.698	43.698	29.962	2.771
69	1	60	1.65	2.8	0.66	2.746	44.548	30.411	2.834
70	5	10	1.98	2.75	4.1	2.905	45.836	31.719	3.027
71	2	60	2.64	2.5	1.73	2.184	37.771	25.09	2.125
72	4	80	2.64	2.75	7.65	3.75	51.853	37.815	4.037
73	1	70	0.99	2.9	0.59	2.164	38.293	24.925	2.104
74	4	10	0.99	2.6	1.26	1.402	23.479	14.228	1.039

(continued on next page)

Table A2 (*continued*)

No.	ML	TL	SD	FD	h_m (mm)	R_s	σ_θ	JRC_m	Z_{2s}
75	5	70	1.65	2.55	4.46	1.727	30.72	19.593	1.518
76	1	30	1.32	2.75	0.3	2.099	37.124	24.212	2.019
77	9	30	0.66	2.5	1.69	1.126	13.195	6.481	0.532
78	2	20	0.33	2.55	0.21	1.044	7.838	1.421	0.304
79	9	50	1.98	2.8	9.04	3.212	48.364	34.098	3.4
80	9	20	1.65	2.75	5.24	2.496	41.951	28.194	2.523
81	2	90	2.97	2.7	4.48	3.736	51.684	37.748	4.023

expression. The Pearson correlation coefficient (PCC) is calculated by using the PCC method to analyse the fracture roughness parameters under the different settings in Table A.2. Fig. 9 shows a scattering diagram of the Pearson correlation coefficient matrix. The diagonal bar diagonally shows the distributions of the parameters R_s, σ_θ, JRC_m, and Z_{2s}, while the remaining subgraphs show the correlation between two specific parameters. The correlations between the characterization parameters are extremely strong, with correlation coefficients ranging from 0.95064 to 0.99795, indicating that each characterization parameter is closely related and can reflect the roughness of the crevasse. The mean correlations between the fracture roughness characterization parameters R_s, σ_θ, JRC_m, and Z_{2s} are 0.97172, 0.97255, 0.98212, and 0.98283, respectively, with Z_{2s} having the highest correlation with the other evaluation index values. Z_{2s} is taken as the most critical factor, as the magnitude and variation trends of these three parameters are more or less the same, so it is reasonable to assume that Z_{2s} alone can represent their overall impact.

The ratio of EHA to h_m during seepage is expressed by C_h, and the proportion of effective transport channels in the fracture aperture is expressed. Fig. 10 shows the relationship between C_h and Z_{2s} under different values of Q_m. A negative correlation between C_h and Z_{2s} is observed, and the change rate of C_h decreases with increasing Z_{2s}. This phenomenon shows that the increase in the roughness of the fracture leads to more vortex and energy consumption in the seepage process, and the EHA decreases accordingly. As Z_{2s} continues to increase, the eddy current in the seepage process tends to stabilize, and the change rate of C_h decreases. With the increase in Q_m, the C_h value decreases, meaning that the ratio of fluid efficient transport space to fracture aperture decreases, validating the nonlinear seepage behaviour in the rough fracture induced by the increase in the internal friction and local energy loss mentioned in Section 2.3.

Table 1 shows the C_h relations under the corresponding conditions fitted with logarithmic functions. Using Q_m and Z_{2s} as independent variables, a custom function $y = ax_1^b(cln(x_2) + d)$ is established to fit the relationship between Z_{2s} and C_h at different Q_m values (Eq. (9)). $R^2 = 0.7004$ for Eq. (9).

$$C_h = 0.0982Q_m^{-0.253}(-0.142\ln(Z_{2s}) + 0.5225) \tag{9}$$

The relationship between EHA and Z_{2s} and h_m is established by the above analysis. h_m and EHA can be approximately expressed using linear functions, and Z_{2s} and EHA can be expressed using logarithmic functions. Based on Eq. (9), Q_m, Z_{2s}, and h_m are selected as independent variables, and the EHA relationship (Eq. (10)) for different Q_m is fitted. $R^2 = 0.9091$ for Eq. (10).

$$h_e = 0.1773Q_m^{-0.2635}(-0.3071\ln(Z_{2s}) + 1.4287)(0.1333h_m + 0.0946) \tag{10}$$

To verify the accuracy of the established formula, Eq. (10) is used to predict the simulation results when $Q_m = 2.0 \times 10^{-4}$, 4.0×10^{-4}, and 6.0×10^{-4} m^3/s, and the predicted results are compared to the theoretical calculations, as shown in Fig. 11. Fig. 11 shows that the horizontal coordinates represent the practical values, the longitudinal coordinates represent the model predictions, and the better the distributions of the data points along the diagonal lines, the more accurate the data projections are. Fig. 11 shows that, in most cases, there are small differences between the EHA predicted by regression-based analytical models and the EHA derived from numerical simulations. The linear relationship between the practical value and the predicted value is satisfied with a linear formula of $y = 0.8929x$-0.0924 and a prediction accuracy of 0.8851, indicating that the overall model can be highly predictive. To date, indoor fracture seepage tests are generally conducted at low flow rates, and there are few tests at high flow rates. For this reason, to further validate the accuracy of the regression model, predictions were made based on 16 sets of data from the literature (Xiao and Zhao, 2019). The results obtained by the regression model are shown in Table A.4. The average relative error between the target value and the output value is calculated to be 8.8%. Fig. 12 shows that the target (practical EHA) is well described by the output (predicted EHA) and that the regression model exhibits high prediction accuracy.

The equivalent hydraulic aperture EHA is quantitatively characterized above using the fractured 3D roughness index Z_{2s}, average mechanical fracture aperture h_m, and volumetric flow Q_m. However, these parameters often play dominant roles in the actual seepage process. The mechanical mechanism of the actual seepage process in rock fractures is complicated by the coupling between fluid flow and rock fracture. If the equivalent hydraulic aperture of rough fracture seepage can be predicted without knowing the details, this is of great significance to the field of hydromechanics. It is well known that the characteristic parameters of rock fractures can be assessed from geological data and that injection flows can be measured. Therefore, the neural network algorithm can be used to set up the prediction model of the equivalent hydraulic aperture in the fracture seepage process. This method can consider the influences of random error and system error and obtain more accurate results than explicit equations. When the neural network is trained, the input parameters are characterized by the statistical parameters, and the global search ability of the algorithm is optimized by the SSA algorithm. The accuracy and efficiency of the algorithm are greatly improved, and the prediction results of the trained neural network are closer to the real values. More details are provided in the next section.

3.2.2. SSA–BP-based model

The back propagation (BP) algorithm is one of the most important algorithms in artificial neural networks, but it has some drawbacks, such as a slow learning convergence speed, no guarantee of convergence to the global minimum point, and an uncertain network structure (Su and Deng, 2003; Jiang et al., 2022). The genetic neural network algorithm (GA–BP) optimizes the stability of the BP algorithm by optimizing the weights and thresholds, but there are still some problems, such as low efficiency, a long solution time, and premature convergence (Fang et al., 2018; Qian et al., 2019). On this basis, Xue and Shen (2020) and other researchers have proposed a new swarm intelligence optimization algorithm—the SSA—which has the characteristics of fast convergence, high search accuracy, stability, and robustness. The SSA has more obvious advantages than existing algorithms, and it has been applied and validated in different fields (Lyu et al., 2021; Jiang, et al., 2021).

A sparrow colony can be divided into discoverers and followers. The discoverers are responsible for searching for food for the population; the followers follow the discoverers to search for food. The identity of the two is dynamic, but the proportion of the population is constant. The discoverer location update formula is as follows:

Table A3
Simulation results of the pressure difference and equivalent hydraulic aperture of water seepage.

No.	$Q_m = 0.0001$ m^3/s		$Q_m = 0.0002$ m^3/s		$Q_m = 0.0003$ m^3/s		$Q_m = 0.0004$ m^3/s		$Q_m = 0.0005$ m^3/s		$Q_m = 0.0006$ m^3/s		$Q_m = 0.0007$ m^3/s		Forchheimer	
	ΔP (Pa)	h_e (mm)	ΔP (Pa)	h_e (mm)	ΔP (Pa)	h_e (mm)	ΔP (Pa)	h_e (mm)	ΔP (Pa)	h_e (mm)	ΔP (Pa)	h_e (mm)	ΔP (Pa)	h_e (mm)	A ($\times 10^6$)	B ($\times 10^{11}$)
1	3467	0.88	11,460	0.75	22,582	0.68	37,400	0.64	56,056	0.6	77,820	0.57	103,240	0.55	21.1	1.81
2	39	3.96	128	3.35	231	3.15	416	2.85	573	2.76	864	2.55	1045	2.52	0.24	0.02
3	10,186	0.62	34,946	0.52	68,877	0.47	113,892	0.44	171,100	0.41	236,838	0.39	314,420	0.38	64.73	5.5
4	420	1.79	1258	1.56	2332	1.46	3636	1.38	5356	1.31	7134	1.26	9419	1.21	3.49	0.14
5	80,163	0.31	282,440	0.26	477,490	0.25	865,520	0.22	1,170,500	0.22	1,749,240	0.2	2,163,200	0.2	660.6	37.58
6	1914	1.08	5986	0.93	11,928	0.85	19,812	0.79	29,624	0.74	41,478	0.7	55,117	0.67	10.33	0.98
7	4761	0.8	16,342	0.66	32,277	0.61	53,484	0.56	80,272	0.53	111,426	0.51	147,810	0.48	29.71	2.6
11	345	1.91	1054	1.66	2034	1.52	3308	1.43	4886	1.35	6762	1.29	8917	1.24	2.27	0.15
15	4623	0.8	14,584	0.69	29,479	0.63	49,168	0.58	74,053	0.55	103,752	0.52	138,680	0.49	22.92	2.5
16	317	1.96	1078	1.65	2153	1.5	3676	1.38	5503	1.3	7794	1.23	10,274	1.18	1.59	0.19
17	3298	0.9	10,072	0.78	19,876	0.71	32,784	0.66	48,886	0.63	68,136	0.6	90,452	0.57	18.76	1.58
18	86,730	0.3	279,560	0.26	554,750	0.24	920,080	0.22	1,380,400	0.21	1,921,560	0.2	2,555,400	0.19	495.4	45.12
19	2470	0.99	9020	0.81	19,275	0.72	33,320	0.66	51,084	0.62	72,900	0.58	98,172	0.56	6.9	1.91
21	1138	1.28	3604	1.1	7189	1	11,768	0.93	17,670	0.88	24,492	0.84	32,741	0.8	6.62	0.57
22	127,450	0.27	382,180	0.23	747,390	0.21	1,226,520	0.2	1,824,600	0.19	2,533,020	0.18	3,358,300	0.17	755.5	57.77
23	78	3.13	246	2.69	515	2.41	812	2.28	1312	2.09	1698	2.04	2426	1.91	0.43	0.04
24	49,775	0.36	147,100	0.32	291,650	0.29	482,760	0.27	721,180	0.26	1,006,980	0.24	1,340,000	0.23	264.1	23.57
25	95	2.93	328	2.45	640	1.55	1136	2.04	1637	1.14	2424	1.81	3073	0.92	0.44	0.06
26	174	2.4	522	2.1	978	1.35	1524	1.85	2247	1.02	3006	1.69	3982	0.84	1.41	0.06
27	873	1.4	2794	1.2	5562	1.09	9108	1.02	13,655	0.96	18,942	0.91	25,279	0.87	5.17	0.44
28	828	1.43	3124	1.15	6144	1.05	10,168	0.98	15,403	0.92	21,132	0.88	28,113	0.84	5.82	0.49
29	1601	1.14	5816	0.94	11,266	0.86	18,512	0.8	27,812	0.76	38,088	0.72	50,381	0.69	11.88	0.86
31	706	1.5	2340	1.27	4644	1.16	7640	1.08	11,476	1.01	15,900	0.97	21,182	0.93	4.3	0.37
32	1356	1.21	4368	1.03	8799	0.94	14,496	0.87	21,822	0.82	30,384	0.78	40,668	0.74	7.44	0.72
34	52	3.59	178	3	321	2.82	596	2.53	806	2.46	1254	2.26	1486	2.24	0.29	0.03
35	54,496	0.35	187,420	0.29	380,290	0.27	637,880	0.25	964,940	0.23	1,351,380	0.22	1,805,200	0.21	279.5	32.88
36	1502	1.17	4650	1.01	9404	0.91	15,620	0.85	23,423	0.8	32,910	0.76	43,942	0.73	7.45	0.79
37	14,356	0.55	45,494	0.47	90,630	0.43	150,748	0.4	226,530	0.38	315,762	0.36	420,610	0.34	78.07	7.47
38	334	1.93	1056	1.66	1996	1.53	3312	1.43	4806	1.36	6768	1.29	8725	1.24	2.28	0.15
39	666	1.53	2006	1.34	4002	1.22	6492	1.14	9745	1.07	13,458	1.02	18,062	0.98	3.83	0.31
40	546	1.64	1768	1.4	3506	1.27	5696	1.19	8605	1.12	11,784	1.07	15,821	1.02	3.44	0.27
41	333	1.93	1004	1.68	1924	1.55	3048	1.47	4551	1.38	6132	1.33	8160	1.27	2.42	0.13
42	76	3.15	240	2.71	488	2.45	800	2.29	1213	2.15	1680	2.05	2265	1.95	0.4	0.04
43	118	2.73	344	2.41	635	2.25	1008	2.12	1491	2	1992	1.93	2628	1.86	0.92	0.04
44	391	1.83	1304	1.54	2619	1.4	4288	1.31	6518	1.23	8952	1.17	12,057	1.12	2.32	0.21
46	510	1.68	1542	1.46	2981	1.34	4844	1.26	7152	1.19	9906	1.13	13,065	1.09	3.31	0.22
47	746	1.48	2228	1.29	4354	1.18	7096	1.11	10,477	1.05	14,604	1	19,297	0.95	4.54	0.33
48	380	1.85	1200	1.59	2367	1.45	3840	1.36	5774	1.28	7920	1.22	10,601	1.17	2.4	0.18
50	8275	0.66	27,438	0.56	54,243	0.51	89,916	0.47	134,920	0.45	187,434	0.43	248,910	0.41	49.59	4.38
51	4119	0.66	10,954	0.76	26,997	0.51	35,348	0.65	45,667	0.45	73,182	0.58	99,483	0.18	21.17	1.68
52	193	2.32	1874	1.37	1445	1.71	7268	1.1	3845	1.46	16,182	0.96	7357	1.32	0.57	0.44
53	12,148	0.58	35,346	0.51	69,649	0.47	114,052	0.44	169,270	0.41	236,118	0.39	313,520	0.38	68.33	5.42
55	817	1.43	2772	1.2	5474	1.1	9144	1.02	13,677	0.96	19,116	0.91	25,262	0.87	4.86	0.45
56	1228	1.25	4234	1.04	8648	0.94	14,548	0.87	22,037	0.82	30,942	0.77	41,386	0.74	5.97	0.76
57	1228	1.25	4234	1.04	8648	0.94	14,548	0.87	22,037	0.82	30,942	0.77	41,386	0.74	5.97	0.76
58	146	2.54	500	2.13	1026	1.91	1800	1.75	2696	1.64	3900	1.55	5136	1.48	0.5	0.1
59	110	2.79	394	2.3	799	2.08	1428	1.89	2086	1.79	3102	1.67	3986	1.62	0.37	0.08
60	95	2.94	298	2.53	640	2.24	1076	2.07	1663	1.93	2334	1.83	3197	1.74	0.29	0.06
62	2193	1.03	6690	0.9	13,445	0.81	22,260	0.76	33,285	0.71	46,710	0.68	62,264	0.65	11.25	1.11
63	9044	0.64	27,618	0.56	55,196	0.51	91,716	0.47	137,380	0.44	192,294	0.42	256,140	0.4	46.89	4.56
66	383	1.84	1350	1.53	2730	1.38	4540	1.28	6966	1.2	9570	1.15	12,912	1.09	2.15	0.23
67	152	2.51	486	2.15	896	2	1532	1.84	2169	1.77	3138	1.66	3931	1.62	1.03	0.07
68	256,590	0.21	764,400	0.18	1,478,700	0.17	2,409,600	0.16	3,569,600	0.15	4,935,600	0.14	6,525,100	0.14	1620	110.1
69	63,883	0.33	191,180	0.29	376,010	0.27	618,280	0.25	920,890	0.24	1,281,300	0.22	1,701,200	0.21	366.1	29.49
70	571	1.61	1796	1.39	3719	1.25	6232	1.15	9535	1.08	13,308	1.03	18,025	0.98	2.38	0.33
71	7226	0.69	23,066	0.59	46,661	0.54	77,972	0.5	117,340	0.47	164,718	0.44	219,900	0.42	35.73	3.98
72	182	2.36	600	2	1248	1.79	2080	1.66	3193	1.55	4440	1.48	6028	1.41	0.8	0.11
74	6647	0.71	20,454	0.62	40,612	0.56	67,468	0.52	101,000	0.49	141,042	0.47	187,550	0.45	35.87	3.32
75	390	1.83	1254	1.56	2440	1.43	4028	1.34	5950	1.26	8322	1.2	10,925	1.15	2.47	0.19
76	294,330	0.2	861,000	0.18	1,670,400	0.16	2,719,600	0.15	4,024,900	0.14	5,575,800	0.14	7,378,000	0.13	1811	124.7
77	3035	0.92	9008	0.81	17,917	0.74	29,536	0.69	44,120	0.65	61,584	0.62	82,060	0.59	16.24	1.44

Table A4
Characteristics of fracture and corresponding EHA.

No.	h_m (mm)	R_s	$Q_m = 0.0002\ m^3/s$		$Q_m = 0.0003\ m^3/s$		$Q_m = 0.0004\ m^3/s$		$Q_m = 0.0005\ m^3/s$	
			target	output	target	output	target	output	target	output
1	1.92	1.529	0.95	0.91	0.83	0.76	0.75	0.63	0.69	0.60
2	1.6	0.876	0.94	0.91	0.83	0.76	0.76	0.63	0.70	0.59
3	2.19	1.669	1.08	0.99	0.94	0.82	0.85	0.68	0.79	0.65
4	1.58	2.312	0.72	0.72	0.63	0.60	0.58	0.50	0.53	0.47
5	1.04	1.997	0.59	0.57	0.52	0.47	0.47	0.40	0.44	0.37
6	1.01	1.175	0.56	0.63	0.49	0.53	0.45	0.44	0.41	0.42
7	1.25	1.793	0.62	0.66	0.54	0.55	0.49	0.45	0.45	0.43
8	1.15	1.374	0.59	0.66	0.51	0.55	0.47	0.46	0.43	0.43
9	1.61	1.488	0.83	0.81	0.73	0.68	0.66	0.56	0.61	0.53
10	1.66	1.019	0.95	0.90	0.84	0.75	0.77	0.63	0.71	0.59
11	1.31	1.196	0.70	0.74	0.61	0.62	0.55	0.52	0.51	0.49
12	1.07	1.333	0.54	0.64	0.47	0.53	0.43	0.44	0.39	0.42
13	2.28	1.95	1.07	0.98	0.94	0.82	0.85	0.68	0.78	0.64
14	1.78	1.534	0.87	0.86	0.76	0.72	0.69	0.60	0.64	0.57
15	1.29	1.582	0.66	0.69	0.58	0.57	0.52	0.48	0.48	0.45
16	1.03	0.75	0.65	0.71	0.58	0.59	0.53	0.49	0.49	0.46

$$X_{i,j}^{t+1} = \begin{cases} X_{i,j}^{t}\cdot\exp\left(\dfrac{-i}{\alpha\cdot N}\right), R < S \\ X_{i,j}^{t} + Q\cdot L, R \geqslant S \end{cases} \tag{11}$$

where t is the number of current iterations, N is the maximum number of iterations, $X_{i,j}$ denotes the position of the i-th sparrow at time j, α is a random number (0,1], R is a warning value, S is a safety value, Q is a random number conforming to a normal distribution, L is a unit matrix of $1 \times d$, and d is the dimension of the variable to be optimized for the problem.

When $R < S$ indicates that the system is safe, the discoverer can lead the follower on a safe foraging trip; when $R \geq S$ indicates the presence of a predator nearby, the population reacts by moving away from the danger area. The follower location update formula during foraging is as follows:

$$X_{i,j}^{t+1} = \begin{cases} Q\cdot\exp\left(\dfrac{X_w^t - X_{i,j}^t}{i^2}\right), i > n\Big/2 \\ X_p^{t+1} + \left|X_{i,j}^t - X_p^{t+1}\right|\cdot A^T\left(AA^T\right)^{-1}\cdot L, i \leqslant n\Big/2 \end{cases} \tag{12}$$

where X_p is the global optimal position, X_w is the global worst position, and A represents the unit matrix of $1 \times d$. When $i > n/2$, the i-th follower has not obtained food and needs to travel to other areas to search for food and improve fitness.

When the population is threatened by predators, some sparrows switch to vigilantes, making up 10% of the population. The location update formula for vigilantes is as follows:

$$X_{i,j}^{t+1} = \begin{cases} X_b^t + \beta\cdot\left|X_{i,j}^t - X_b^t\right|, f_i > f_g \\ X_{i,j}^t + K\cdot\left[\dfrac{\left|X_{i,j}^t - X_w^t\right|}{(f_i - f_w) + \varepsilon}\right], f_i = f_g \end{cases} \tag{13}$$

where X_b is the global optimal location, β is a step length adjustment parameter subject to a normal distribution random number with a variance of 1 and a mean of 0, K is a random number [-1,1], f_i is the fitness value of the i-th individual, f_g is the global optimal adaptation value, f_w is the worst adaptation value, and ε is a constant to prevent the denominator from being 0. When $f_i > f_g$, the population is vulnerable to predators. When $f_i = f_g$, the sparrow is aware of the danger and needs to move quickly towards the sparrows in other areas.

In this section, an equivalent hydraulic aperture EHA prediction model is established by using the SSA–BP neural network algorithm, which is optimized by the SSA. Six parameters are selected as input parameters, including 5 fracture parameters (h_m, R_s, JRC_m, Z_{2s}, σ_θ) and 1 seepage parameter (Q_m). One neuron in the output layer is the EHA. Learning is based on numerical simulations of 427 data sets randomly divided into training sets (75%, 320 sets) and test sets (25%, 107 sets). The SSA–BP neural network parameters used in this study are shown in Table 2. The SSA–BP neural network algorithm flow is shown in Fig. 13, with the following steps:

(1) First, the topological structure of the BP neural network and input and output variables are determined, and the algorithm stop condition is set to the maximum training error value $e_{train} \leq 10^{-4}$.
(2) The weights and thresholds of the BP neural networks are initialized, and the parameters associated with the SSA are defined, such as N, β, S, etc.
(3) The adaptive value of the sparrow population is calculated and sequenced from the minimum to the maximum values to find the optimal and worst values. The initial adaptive value is the training error of the BP neural network random initial value.
(4) Eqs. (11)–(13) are used to update the locations of discoverers, followers, and vigilantes, respectively.
(5) The optimal values of the iteration sparrow position are obtained, and the position is updated if the new position is better than the optimal value of the previous iteration; otherwise, location updates are not performed.
(6) If N reaches the maximum number of iterations 100, the search is stopped, and the global optimal value and the best fitness value are output; otherwise, steps 3–5 are repeated.
(7) The initial weight and threshold of the BP neural network are the global optimal corresponding parameters of the SSA, and the network is trained by input training and a test sample set.

To prevent overfitting of the model, 1 to 15 layers of the implicit neural network are selected and predicted using the BP, GA–BP, and SSA–BP models, and the corresponding root mean squared error (RMSE) values are recorded. RMSE is the square root of the ratio of the difference between the predictive value of the parameter and the true value of the parameter and the ratio of the total number of samples to n. The accuracy of predictive models is well evaluated by RMSE, which is expressed as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2} \tag{14}$$

Fig. 14 shows the RMSE for three prediction models with different implied layers, and overall, the RMSE for all three models is under 0.12. The SSA–BP model has a lower RMSE than the GA–BP and BP models. The optimal number of hidden layers of the neural network in the three models is between 5 and 9 layers. When the number of hidden layers of the neural network exceeds 9 layers, the prediction error tends to increase, and an overfitting phenomenon occurs. Fig. 15 shows an evaluation of the correlation coefficient (R) of three neural networks when the number of hidden layers is 9, which is defined as follows:

$$R = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2 \cdot \sum_{i=1}^{n}(y_i - \overline{y})^2}} \tag{24}$$

where x_i and y_i are two variables representing the predicted and target values, respectively, n is the total sample size, and ^{-}x and ^{-}y are the average values of the two variables.

As seen in Fig. 15, all involved R values approach 1, indicating that the target (practical EHA) is well described by the output (predicted EHA) and that the SSA–BP model exhibits high prediction accuracy; these findings suggest that SSA optimization significantly improves the BP neural network inversion capabilities.

To examine the degree of accuracy of the predictive model, the distribution of the relative error between the practical and predicted EHAs is shown in Fig. 16 using a box plot, where the relative error between 11.401% and −11.8656% accounts for 93.91% of all sample points, among which only 26 are outliers (shown as “+”). We define two variables: upper adjacent (UA) and lower adjacent (LA), with UA $= q_3 + w(q_3\text{-}q_1) = 7.146$ and LA $= q_1\text{-}w(q_3\text{-}q_1) = -8.3936$, where w is the maximum whisker length of default value 1.5 and $q_3 = 1.4626$ and $q_1 = -2.5139$ are the 75th and 25th percentiles, respectively. A percentile is the threshold value where a given percentage of sample points in a set of sample points fall. The outliers are points with a relative error larger than UA and less than LA. Fig. 16 shows the statistical distribution of the relative error between the practical and predicted EHAs at different velocities. At $Q_m = 0.0001$ m^3/s and 0.0007 m^3/s, the IQR values of the prediction model are 5.3476 and 5.1519, respectively, which are slightly larger than the IQR range of 2.9163–4.3478 at $Q_m = 0.0002$–0.0006 m^3/s, where IQR $= q_3\text{-}q_1$ is the interquartile range. IQR represents the concentration distribution of relative errors. Because the inlet velocity values of the first two are at the edge of the model-trained data range, the distribution is more dispersed. It is shown that training with more widely used data can produce better predictive models with improved accuracy.

4. Conclusion

Due to the roughness of a fracture wall, a fracture aperture does not fully contribute to fluid transport; effective fluid transport is concentrated only in the central part of the fracture, which can be quantitatively assessed by the EHA.

With increasing flow rate, the effective flat-flow zone narrows, and the inertia increases. Additionally, the friction resistance and local energy loss in the fluid increase, which leads to nonlinear seepage behaviour in the rough fracture.

A predictive mathematical model between an EHA and Z_{2s}, h_m and Q_m is established. The results show that the model can be highly predictive in the EHA under $Q_m = 1.0 \times 10^{-4}$-7.0×10^{-4} m^3/s.

An EHA prediction model based on the SSA–BP neural network is constructed considering fracture geometry and flow parameters. It is found that the wider the range of data used for training, the more accurate the model is.

In the past, it was necessary to manually modify the parameters when constructing numerical models. The automatic simulation program proposed in this paper greatly shortens the manual time, improves the simulation efficiency, and provides a reliable method for systematic large-scale numerical simulation experiments. However, it is difficult to consider the influences of the heterogeneity, anisotropy, and random distribution of the fractures in an actual rock mass on the hydraulic characteristics by the numerical calculation method to a certain extent, which is a difficult problem to solve in the field of fracture seepage. Moreover, to control the variables and simplify the research, a fracture model with the same size is used in this paper, and the influences of the size on the seepage properties have not been discussed. These issues will be the subject of our team’s next in-depth study.

CRediT authorship contribution statement

Weijie Zhang: Investigation, Methodology, Software, Funding acquisition, Writing – original draft. **Ziyu Peng:** Data curation, Software, Formal analysis, Writing – original draft. **Chenghao Han:** Investigation, Visualization, Software, Project administration, Writing - review & editing. **Shaojie Chen:** Methodology, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Appendix A

(SEE Tables A1-A4).

References

Arianfar, A., Ramezanzadeh, A., Khalili, M., 2021. Numerical study of nonlinear fluid flow behavior in natural fractures adjacent to porous medium. J. Pet. Sci. Eng. 204 https://doi.org/10.1016/j.petrol.2021.108710.

Boussinesq, J., 1868. Mémoire sur l’influence de frottement dans les mouvements réguliers des fluides. J. Math. Pureset Appl. 13, 377–424.

Brown, S.R., 1987. Fluid flow through rock joints: the effect of surface roughness. J. Geophys. Res.: Solid Earth. 92 (B2), 1337–1347. https://doi.org/10.1029/JB092iB02p01337.

Brown, S.R., 1995. Simple mathematical model of a rough fracture. J. Geophys. Res.: Solid Earth. 100 (B4), 5941–5952. https://doi.org/10.1016/0148-9062(96)83910-6.

Belem, T., Homand-Etienne, F., Souley, M., 2000. Quantitative parameters for rock joint surface roughness. Rock Mech Rock Eng. 33 (4), 217–242. https://doi.org/10.1007/s006030070001.

Brush, D.J., Thomson, N.R., 2003. Fluid flow in synthetic rough-walled fractures: Navier-Stokes, Stokes, and local cubic law simulations. Water Resour. Res. 39, 4. https://doi.org/10.1029/2002WR001346.

Briggs, S.A., 2014. Impact of Single Fracture Roughness on the Flow, Transport and Development of Biofilms. University of Toronto, Canada.

Briggs, S., Karney, B.W., Sleep, B.E., 2014. Numerical modelling of flow and transport in rough fractures. J. Rock Mech. Geotech. 6 (6), 535–545. https://doi.org/10.1016/j.jrmge.2014.10.004.

Briggs, S., Karney, B.W., Sleep, B.E., 2017. Numerical modeling of the effects of roughness on flow and eddy formation in fractures. J. Rock Mech. Geotech. 9 (01), 105–115. https://doi.org/10.1016/j.jrmge.2016.08.004.

Crandall, D., Bromhal, G., Karpyn, Z.T., 2010. Numerical simulations examining the relationship between wall-roughness and fluid flow in rock fractures. Int. J. Rock Mech. Min. 47 (5), 784–796. https://doi.org/10.1016/j.ijrmms.2010.03.015.

Chen, Y.F., Zhou, J.Q., Hu, S.H., Hu, R., Zhou, C.B., 2015. Evaluation of Forchheimer equation coefficients for non-Darcy flow in deformable rough-walled fractures. J. Hydrol. 529, 993–1006. https://doi.org/10.1016/j.jhydrol.2015.09.021.

Develi, K., Babadagli, T., 2015. Experimental and visual analysis of singlephase flow through rough fracture replicas. Int. J. Rock Mech. Min. 73, 139–155. https://doi.org/10.1016/j.ijrmms.2014.11.002.

Dang, W., Wu, W., Konietzky, H., Qian, J., 2019. Effect of shear-induced aperture evolution on fluid flow in rock fractures. Comput. Geotech. 114, 103152-. https://doi.org/10.1016/j.compgeo.2019.103152.

Fang, P., Cai, Z., Zhang, P. 2018. Application of the GA-BP Neural Network in Earthwork Calculation. IOP Conf. Ser.: Mater. Sci. Eng. 301.01. https://doi.org/10.1088/1757-899X/301/1/012130.

Goodman, R.E., 1976. Methods of geological engineering in discontinuous rocks. West Publ. Co, St. Paul, 158–208. https://doi.org/10.1016/0148-9062(76)90566-0.

Glover, P., Matsuki, K., Hikima, R., Hayashi, K., 1998. Synthetic rough fractures in rocks. J. Geophys. Res.: Atmos. 103 (5), 9609–9620. https://doi.org/10.1029/97JB02836.

Ge, Y.F., Tang, H.M., Huang, L., Wang, L.Q., Sun, M.J., Fan, Y.J., 2012. A new representation method for three-dimensional joint roughness coefficient of rock mass

discontinuities. Chin. J. Rock Mech. Eng. 31 (12), 2508–2517. https://doi.org/10.3969/j.issn.1000-6915.2012.12.015.
Hsiung, S.M., Ghosh, A., Ahola, M.P., Chowdhury, A.H., 1993. Assessment of conventional methodologies for joint roughness coefficient determination. Int. J. Rock Mech. Min. Sci. Geomech. Abstr. 30 (7), 825–829. https://doi.org/10.1016/0148-9062(93)90030-H.
Jiang, Y., Ma, Y., Liang, Y.Z., Wang, Y., Li, G.H., Ma, D., 2021. Lung tissue segmentation algorithm: fractional order sparrow search optimization for OTSU. Comput. Sci. 48 (S1), 28–32. https://doi.org/10.11896/jsjkx.200900176.
Jiang, D., Liu, C.H., Ou, Z.H., 2022. Classification Prediction Model Based on BP Neural Network. J. Phys.: Conf. Ser. 2218 (1) https://doi.org/10.1088/1742-6596/2218/1/012040.
Kishida, K., Mgaya, P., Ogura, K., Hosoda, T., 2009. Flow on a single rock fracture in the shear process and the validity of the cubic law examined through experimental results and numerical simulations. Soils Found. 49 (4), 597–610. https://doi.org/10.3208/sandf.49.597.
Lomize, G.M., 1951. Flow in fractured rocks. Gosenergoizdat, Moscow.
Lanaro, F., 2000. A random field model for surface roughness and aperture of rock fractures. Int. J. Rock Mech. Min. Sci. 37 (08), 1195–1210. https://doi.org/10.1016/S1365-1609(00)00052-6.
Lyu, X., Mu, X.D., Zhang, J., 2021. Multi-threshold image segmentation based on improved sparrow search algorithm. Syst. Eng. Electron. 43 (02), 318–327. https://doi.org/10.12305/j.issn.1001-506X.2021.02.05.
Mandelbrot, B.B., 1985. Self-affine fractals and fractal dimension. Phys. Scripta. 32 (04), 257. https://doi.org/10.1088/0031-8949/32/4/001.
Mwetulundila, A.L., Atangana, A., 2020. Applying the Forchheimer equation to model an artificially recharged fractured aquifer. Alex. Eng. J. https://doi.org/10.1016/j.aej.2020.01.026.
Ogilvie, S.R., Isakov, E., Glover, P.W.J., 2006. Fluid flow through rough fractures in rocks. II: A new matching model for rough rock fractures. Earth Planet. Sc. Lett. 241 (3–4), 454–465. https://doi.org/10.1016/j.epsl.2005.11.041.
Power, W.L., Tullis, T.E., 1991. Euclidean and fractal models for the description of rock surface roughness. J. Geophys. Res.: Solid. Earth. 96 (B1), 415–424. https://doi.org/10.1029/90JB02107.
Qian, S.Q., Liu, Y.M., Ye, Y.Q., Xu, G.F., 2019. An enhanced genetic algorithm for constrained knapsack problems in dynamic environments. Nat. Comput. 18 (04), 913–932. https://doi.org/10.1007/s11047-018-09725-3.
Rong, G., Hou, D., Yang, J., Cheng, L., Zhou, C., 2017. Experimental study of flow characteristics in non-mated rock fractures considering 3D definition of fracture surfaces. Eng. Geol. 220, 152–163. https://doi.org/10.1016/j.enggeo.2017.02.005.
Schrauf, T.W., Evans, D.D., 1986. Laboratory Studies of Gas Flow Through a Single Natural Fracture. Water Resour. Res. 22 (7), 1038–1050. https://doi.org/10.1029/WR022i007p01038.
Sun, H.Q., Xie, H.P., 1999. Fractal simulation of the fault surface. J. China Univ. Min. Technol. 28 (5), 433–436. https://doi.org/10.3321/j.issn:1000-1964.1999.05.006.
Su, G.L., Deng, F.P., 2003. On the improving backpropagation algorithms of the neural networks based on MATLAB language: A review. B. Sci. Technol. 02, 130–135. https://doi.org/10.13774/j.cnki.kjtb.2003.02.012.
Song, Y., Liu, B.G., Ren, D.R., Yu, M.Y., Huang, R., 2021. Study on stochastic method for modeling rough joints based on fractal theory. Chin. J. Rock Mech. Eng. 40 (01), 101–112. https://doi.org/10.13722/j.cnki.jrme.2020.0487.
Tsang, Y.W., Witherspoon, P.A., 1983. The dependence of fracture mechanical and fluid flow properties on fracture roughness and sample size. J. Geophys. Res.: Solid. Earth. 88 (B3), 2359–2366. https://doi.org/10.1029/JB088iB03p02359.
Tatone, B.S.A., Grasselli, G., 2012. Quantitative Measurements of Fracture Aperture and Directional Roughness from Rock Cores. Rock Mech. Rock Eng. 45 (4), 619–629. https://doi.org/10.1007/s00603-011-0219-5.
Wang, Y., Su, B.Y., 2002. Research on the behavior of fluid flow in a single fracture and its equivalent hydraulic aperture. Adv. Water Sci. 01, 61–68.
Wang, L.C., Cardenas, M.B., Slottke, D.T., Ketcham, R.A., Sharp, J.M., 2015. Modification of the local cubic law of fracture flow for weak inertia, tortuosity, and roughness. Water Resour. Res. 51 (4), 2064–2080. https://doi.org/10.1002/2014WR015815.
Wang, B., Wang, Y., Niu, Y.L., 2017. Determination on hydraulic aperture of single rough fractures based on standard curves of discrete joint roughness coefficient. Water Resour Pow. 35 (04), 77.
Xie, H., Wang, J.A., Stein, E., 1998. Direct fractal measurement and multifractal properties of fracture surfaces. Phys Lett A. 242 (1-2), 41–50. https://doi.org/10.1016/S0375-9601(98)00098-X.
Xiong, F., Sun, H., Jiang, Q.H., Ye, Z.Y., Xue, D.R., Liu, R.Y., 2018. Theoretical model and experimental verification on non-linear flow at low velocity through rough-walled rock fracture. Rock Soil Mech. 39 (9), 3294–3302. http://ytlx.whrsm.ac.cn/CN/Y2018/V39/I9/3294.
Xiao, F., Zhao, Z., Y., 2019. Evaluation of equivalent hydraulic aperture (EHA) for rough rock fractures. Can. Geotech. J. 56 (10), 1486–1501. https://doi.org/10.1139/cgj-2018-0274.
Xue, J.K., Shen, B., 2020. A novel swarm intelligence optimization approach: sparrow search algorithm. Syst. Sci. Control Eng. 8 (1), 22–34. https://doi.org/10.1080/21642583.2019.1708830.
Xiong, F., Jiang, Q.H., Chen, S.Y., Hu, X.C., 2021. Modeling of coupled Darcy-Forchheimer flow in fractured porous media and its engineering application. C. J. Geotech. Eng. 43 (11), 2037–2045.
Xing, K., Qian, J.Z., Zhao, W.D., Ma, H.C., Ma, L., 2021. Experimental and numerical study for the inertial dependence of non-Darcy coefficient in rough single fractures. J. Hydrol. 603, 127148-. https://doi.org/10.1016/j.jhydrol.2021.127148.
Xu, Q., Yu, Y.Q., 2022. Corrected cubic law for rock fracture seepage and its application. Eng. J. Wuhan Univ. 1–9. http://kns.cnki.net/kcms/detail/42.1675.T.20220415.1920.006.html.
Yang, Z.Y., Di, C.C., Yen, K.C., 2001. The effect of asperity order on the roughness of rock joints. Int. J. Rock Mech. Min. Sci. 38 (5), 745–752. https://doi.org/10.1016/S1365-1609(01)00032-6.
Zhong, J.G., Liang, X., Ren, Z.G., 2007. Statistical determination of the equivalent hydraulic aperture of rock fissures. B. Geol. Sci. Tech. 04, 103–106. https://doi.org/10.3969/j.issn.1000-7849.2007.04.021.
Zhang, Z., Nemcik, J., 2013. Friction factor of water flow through rough rock fractures. Rock Mech. Rock Eng. 46 (5), 1125–1134. https://doi.org/10.1007/s00603-012-0328-9.
Zou, L.C., Lanru Jing, L., Cvetkovic, V., 2015. Roughness decomposition and nonlinear fluid flow in a single rock fracture. Int. J. Rock Mech. Min. Sci. 75, 102–118. https://doi.org/10.1016/j.ijrmms.2015.01.016.
Zhang, W., Dai, B.B., Liu, Z., Zhou, C.Y., 2017. A pore-scale numerical model for non-Darcy fluid flow through rough-walled fractures. Comput. Geotech. 87, 139–148. https://doi.org/10.1016/j.compgeo.2017.02.011.
Zhang, Y., Yin, X.Q., Chen, K., 2018. Laboratory study on the features of the high-speed non-darcy seepage flow in single rough fracture. J. Hebei Univ. Eng. (Nat. Sci. Ed.) 35 (1), 6. https://doi.org/10.3969/j.issn.1673-9469.2018.01.004.